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UNIVERSITY of CAMBRIDGE
International Examinations

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Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Binomial Theorem

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \ldots + \binom{n}{r} a^{n-r} b^r + \ldots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{(n-r)!r!}$.

2. TRIGONOMETRY

Identities

$\sin^2 A + \cos^2 A = 1.$

$\sec^2 A = 1 + \tan^2 A.$

$\operatorname{cosec}^2 A = 1 + \cot^2 A.$

Formulae for ΔABC

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

$$a^2 = b^2 + c^2 - 2bc \cos A.$$

$$\Delta = \frac{1}{2} bc \sin A.$$

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1 The functions f and g are defined for $x \in \mathbb{R}$ by

$$\mathrm{f} : x \mapsto x^3,$$

$$\mathrm{g} : x \mapsto x + 2.$$

Express each of the following as a composite function, using only f, g, f^{-1} and/or g^{-1} :

(i) $x \mapsto x^3 + 2$, [1]

(ii) $x \mapsto x^3 - 2$, [1]

(iii) $x \mapsto (x + 2)^{\frac{1}{3}}$. [1]

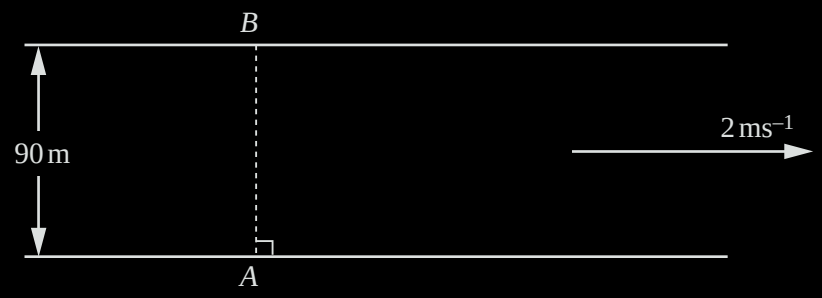
2 Prove the identity

$$\cos x \cot x + \sin x \equiv \operatorname{cosec} x .$$ [4]

3 Evaluate

$$\int_0^{\frac{\pi}{6}} \sin\left(2x + \frac{\pi}{6}\right) \mathrm{d}x.$$ [4]

4



The diagram shows a river 90 m wide, flowing at $2\,\mathrm{ms}^{-1}$ between parallel banks. A ferry travels in a straight line from a point A to a point B directly opposite A. Given that the ferry takes exactly one minute to cross the river, find

(i) the speed of the ferry in still water, [3]

(ii) the angle to the bank at which the ferry must be steered. [2]

5 The straight line $2x + y = 14$ intersects the curve $2x^2 - y^2 = 2xy - 6$ at the points A and B. Show that the length of AB is $24\sqrt{5}$ units. [7]

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6 A curve has equation $y = x^3 + ax + b$, where a and b are constants. The gradient of the curve at the point (2, 7) is 3. Find

(i) the value of a and of b, [5]

(ii) the coordinates of the other point on the curve where the gradient is 3. [2]

7 **(a)** Find the value of m for which the line $y = mx - 3$ is a tangent to the curve $y = x + \frac{1}{x}$ and find the x-coordinate of the point at which this tangent touches the curve. [5]

(b) Find the value of c and of d for which $\{x : -5 < x < 3\}$ is the solution set of $x^2 + cx < d$. [2]

8 Given that $\mathbf{A} = \begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}$, use the inverse matrix of **A** to

(i) solve the simultaneous equations

$$y - 4x + 8 = 0,$$

$$2y - 3x + 1 = 0,$$

(ii) find the matrix **B** such that $\mathbf{BA} = \begin{pmatrix} -2 & 3 \\ 9 & -1 \end{pmatrix}$. [8]

9 **(a)** Express $\left(2 - \sqrt{5}\right)^2 - \frac{8}{3 - \sqrt{5}}$ in the form $p + q\sqrt{5}$, where p and q are integers. [4]

(b) Given that $\frac{a^x}{b^{3-x}} \times \frac{b^y}{\left(a^{y+1}\right)^2} = ab^6$, find the value of x and of y. [4]

10 **(a)** How many different four-digit numbers can be formed from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9 if no digit may be repeated? [2]

(b) In a group of 13 entertainers, 8 are singers and 5 are comedians. A concert is to be given by 5 of these entertainers. In the concert there must be at least 1 comedian and there must be more singers than comedians. Find the number of different ways that the 5 entertainers can be selected. [6]

11 The equation of a curve is $y = xe^{-\frac{x}{2}}$.

(i) Show that $\frac{dy}{dx} = \frac{1}{2}(2 - x)e^{-\frac{x}{2}}$. [3]

(ii) Find an expression for $\frac{d^2y}{dx^2}$. [2]

The curve has a stationary point at M.

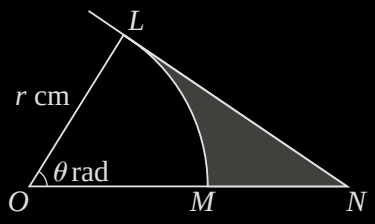
(iii) Find the coordinates of M. [2]

(iv) Determine the nature of the stationary point at M. [2]

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12 Answer only **one** of the following two alternatives.

EITHER



The diagram shows a sector of a circle, centre O and radius r cm. Angle LOM is θ radians. The tangent to the circle at L meets the line through O and M at N. The shaded region shown has perimeter P cm and area A cm^2. Obtain an expression, in terms of r and θ, for

(i) P, [4]

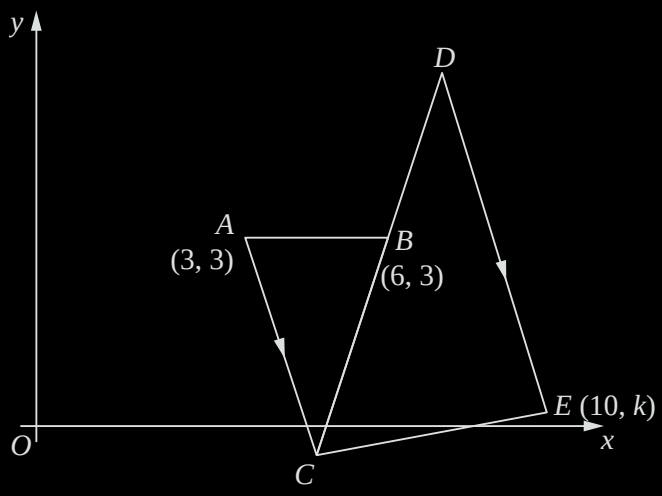
(ii) A. [3]

Given that $\theta = 1.2$ and that $P = 83$, find the value of

(iii) r, [2]

(iv) A. [1]

OR **Solutions to this question by accurate drawing will not be accepted.**



The diagram shows an isosceles triangle ABC in which A is the point (3, 3), B is the point (6, 3) and C lies below the x-axis. Given that the area of triangle ABC is 6 square units,

(i) find the coordinates of C. [3]

The line CB is extended to the point D so that B is the mid-point of CD.

(ii) Find the coordinates of D. [2]

A line is drawn from D, parallel to AC, to the point E (10, k) and C is joined to E.

(iii) Find the value of k. [3]

(iv) Prove that angle CED is **not** a right angle. [2]

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